Relational Algebra
(Operations on Relations & Tables)

Selection

Select tuples where condition c is true: $\sigma_c(R)$

Product

prod_name	prod_price	prod_man
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

$\sigma_{prod_price>20}(Product)$

prod_name	prod_price	prod_man
Powergizmo	$39.99	GizmoWorks
HyperWidget	$203.99	Hyper

Projection

Pick some attributes: $\Pi_{a,\ldots,z}(R)$

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VaporWare	NULL	NULL

$\Pi_{prod_name,prod_man}(Product)$

prod_name	prod_man
Gizmo	GizmoWorks
Powergizmo	GizmoWorks
Widget	WidgetsRUs
HyperWidget	Hyper
VaporWare	NULL

Cross Product

R

a	b
x	1
f	2
z	6

S

c	d
b	5
f	2
h	6

$R \times S$

a	b	c	d
x	1	b	5
x	1	f	2
x	1	h	6
f	2	b	5
f	2	f	2
f	2	h	6
z	6	b	5
z	6	f	2
z	6	h	6

Inner join

Join tables on condition: $R \bowtie_c S$

$$R \bowtie_c S = \sigma_c(R \times S)$$

Inner joins & FKs

Inner join condition is usually on FK relationship.

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VaporWare	NULL	NULL

Manufacturer

m_name	m_address	m_city	m_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Product

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Hyper	1 Mission Dr.	San Francisco	CA

$$Product \bowtie_{p_man=m_name} Manufacturer$$

p_name	p_price	p_man	m_address	m_city	m_state
Gizmo	$19.99	GizmoWorks	123 Gizmo St.	Houston	TX
Powergizmo	$39.99	GizmoWorks	123 Gizmo St.	Huston	TX
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Hyper	1 Mission Dr.	San Francisco	CA

$$Product \triangleright_{p_man=m_name} Manufacturer$$

p_name	p_price	p_man
VaporWare	NULL	NULL

Product

name	price	man
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Manufacturer

name	address	city	state
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WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

$$Product \bowtie_{man=name} Manufacturer$$

name	price	man	address	city	state
Gizmo	$19.99	GizmoWorks	123 Gizmo St.	Houston	TX
Powergizmo	$39.99	GizmoWorks	123 Gizmo St.	Huston	TX
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Manufacturer

name	address	city	state
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WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

$$Product \bowtie_{Product.man=Manufacturer.name} Manufacturer$$

name	price	man	address	city	state
Gizmo	$19.99	GizmoWorks	123 Gizmo St.	Houston	TX
Powergizmo	$39.99	GizmoWorks	123 Gizmo St.	Huston	TX
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Renaming

Rename attribute b to a: $\rho_{a/b}(R)$

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HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

$\rho_{p_name/name}(Product)$

p_name	price	man
Gizmo	$19.99	GizmoWorks
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VaporWare	NULL	NULL

Product

p_name	p_price	p_man

Manufacturer

m_name	m_address	m_city	m_state

$$\{m.m_name \mid Manufacturer(m) \wedge \exists p1 \neg\exists p2 (Product(p1) \wedge Product(p2) \wedge p1.p_man = m.m_name \wedge p2.p_man = m.m_name \wedge p1.p_name \neq p2.p_name\}$$

RELATIONS ARE SETS
TABLES ARE MULTISETS

Union

R

a	b
x	1
f	2
z	6

S

a	b
b	5
f	2
h	6

$R \cup S$

a	b
x	1
f	2
z	6
b	5
h	6

Intersection

R

a	b
x	1
f	2
z	6

S

a	b
b	5
f	2
h	6

$R \cap S$

a	b
f	2

Difference

R

a	b
x	1
f	2
z	6

S

a	b
b	5
f	2
h	6

$R - S$

a	b
x	1
z	6

Two ways to think about multisets

Tuple
(1, a)
(1, a)
(1, b)
(2, c)
(2, c)
(2, c)
(1, d)
(1, d)

=

Tuple	λ
(1, a)	2
(1, b)	1
(2, c)	3
(1, d)	2

Multiset union

X

Tuple	λ
(1, a)	2
(1, b)	0
(2, c)	3
(1, d)	0

$\cup$

Y

Tuple	λ
(1, a)	5
(1, b)	1
(2, c)	2
(1, d)	2

$=$

Z

Tuple	λ
(1, a)	7
(1, b)	1
(2, c)	5
(1, d)	2

$$\lambda(Z) = \lambda(X) + \lambda(Y)$$

Multiset intersection

X

Tuple	λ
(1, a)	2
(1, b)	0
(2, c)	3
(1, d)	0

$\cap$

Y

Tuple	λ
(1, a)	5
(1, b)	1
(2, c)	2
(1, d)	2

$=$

Z

Tuple	λ
(1, a)	2
(1, b)	0
(2, c)	2
(1, d)	0

$$\boldsymbol{\lambda}(\boldsymbol{Z}) = min\big(\boldsymbol{\lambda}(\boldsymbol{X}), \boldsymbol{\lambda}(\boldsymbol{Y})\big)$$

Multiset difference

X

Tuple	λ
(1, a)	2
(1, b)	0
(2, c)	3
(1, d)	0

—

Y

Tuple	λ
(1, a)	5
(1, b)	1
(2, c)	2
(1, d)	2

=

Z

Tuple	λ
(1, a)	0
(1, b)	0
(2, c)	1
(1, d)	0

$$\lambda(Z) = \lambda(X) - \lambda(Y)$$

Relational algebra

- Operations on relations:
  - Projection, selection, renaming
  - Joins (these and more)
  - Set operations
  - Aggregation
  - Transitive closures

- Equality rules